

IAD-Risk: Risk-Aware Identity-Agenda Disentanglement for Scholarly Work Deduplication

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Abstract

Scholarly work deduplication must decide whether two records describe the same work, not whether they merely share a research agenda. This distinction is fragile because scientific representation and pair-classification models can assign high semantic scores to distinct papers from the same task, benchmark, or citation community. We study this failure mode as *identity-agenda confusion* and propose IAD-Risk, a risk-aware framework that treats agenda relatedness as possible false-merge risk rather than direct identity evidence. IAD-Risk combines stratified data provenance, identity and agenda relation heads, agenda-non-identity hard negatives, and false-merge risk gating to block risky automatic pair merges before clustering. We evaluate the framework with IAD-Bench, a benchmark contract that separates DeepMatcher gold labels, SciRepEval/SciDocs-style proxy labels, and OpenAlex/OpenCitations silver hard negatives. In the Open-v2 evidence snapshot, the result rows are scope-bounded mechanism evidence rather than a same-scope comparative ranking: single-space scientific representation baselines show HNFMR 0.790–0.999 on the full pair scope, whereas transformer-backed IAD-Risk variants report same-work F1=0.980 and HNFMR=0.000 on the held-out test scope. The results support a conservative pair-level conclusion: identity-agenda disentanglement can control risky automatic merges in the reported setting, while cluster-level quality claims require cluster artifacts and broader manual or source-heldout validation before broad method-ranking claims.

Keywords: scholarly entity matching, work deduplication, identity-agenda disentanglement, hard-negative false-merge rate, false-merge risk, provenance-aware evaluation, scientific document representation

1. Introduction

Scholarly knowledge systems rely on entity matching to decide whether multiple records describe the same scientific work. Digital libraries, literature review tools, citation graphs, recommender systems, and evidence synthesis pipelines all require stable work identities because a false merge can combine evidence from different papers and distort downstream analysis. This paper focuses on scholarly work deduplication: given two scholarly records with titles, abstracts, authors, venues, identifiers, topics, and citations, the system must decide whether an automatic merge is safe. The operational target is therefore not only to find likely duplicates, but also to avoid false merges that contaminate downstream scholarly records.

The central technical challenge is that agenda evidence can resemble identity evidence. Two papers can share a task, dataset, method family, citation neighborhood, or OpenAlex topic while making different contributions. Scientific representation models such as SPECTER and SciNCL are valuable because they bring related papers close in embedding space [1, 2]; that behavior supports retrieval and candidate generation, but becomes unsafe when semantic closeness is used as merge evidence. This failure is particularly important in scholarly settings because false merges are harder to repair than missed duplicates: once distinct papers are clustered as one work, citation counts, recommendation signals, and review evidence can be contaminated. Merge safety therefore requires explicit attention to hard-negative cases where two records are related but non-identical.

We formulate this failure mode as identity-agenda confusion. Identity refers to whether two records denote the same scholarly work. Agenda refers to whether two records discuss the same research problem, benchmark, method family, or citation community. Existing entity matching methods [3–5] typically optimize pair classification, while scholarly document representation methods optimize semantic relatedness. Neither objective alone makes the desired decision boundary explicit: same-agenda non-identities should increase false-merge risk rather than merge confidence. The paper therefore treats identity-agenda confusion as a merge-safety failure in scholarly deduplication, not as an ordinary semantic-similarity error.

1.1. Motivating Failure Case

A simple failure case makes the distinction concrete. Consider two scholarly records that share the same task or benchmark and use similar termi-

nology, but report a different contribution, model variant, or experimental claim. A scientific representation model may assign high semantic similarity because the records are close in topic space. For work deduplication, however, the safe decision is to avoid an unsafe automatic merge: without DOI agreement, version evidence, or other same-work identity evidence, the pair should be deferred or blocked rather than merged. The point is that agenda relatedness is not identity evidence; it is a risk condition that should be handled explicitly by HNFMR-style evaluation.

Table 1: Motivating failure case for identity-agenda confusion. The example is an illustrative failure case, not a prevalence estimate.

Observed pair signal	Agenda interpretation	Merge-safety interpretation
Same task or benchmark with overlapping technical terms.	The records are related and may receive high semantic similarity.	Relatedness alone is insufficient for automatic same-work identity.
Different contribution, model variant, or experimental claim.	The papers can still belong to one research agenda.	The pair requires same-work identity evidence before merging.
No shared DOI, version marker, or clear publication lineage.	A relatedness model may still rank the pair highly.	Treating the pair as an automatic merge is an unsafe automatic merge and contributes to HNFMR.

We propose IAD-Risk, a risk-aware identity-agenda disentanglement framework for scholarly work deduplication. IAD-Risk learns separate relation signals for same-work identity, same-agenda relatedness, and agenda-non-identity hard negatives, then derives a false-merge risk score from these signals. A pair is eligible for automatic merging only when identity evidence is sufficiently strong and agenda-non-identity risk remains below a merge threshold. This selective decision rule turns deduplication from a high-recall similarity problem into a selective safety problem with explicit rejection thresholds.

The paper makes three contributions. First, it defines identity-agenda confusion as a targeted failure mode in scholarly entity matching and introduces hard-negative false-merge rate as a safety metric. Second, it presents IAD-Bench, a provenance-aware benchmark contract that keeps gold, proxy, silver, and human-reviewed labels separate. Third, it evaluates IAD-Risk un-

der the same pair contract used by representation and entity-matching baselines, while keeping clustering and artifact-audit interfaces as reproducible extensions rather than unsupported primary evidence. The current evidence supports targeted false-merge suppression under stratified evaluation; it does not support a broad method-ranking claim or a claim that silver labels replace human gold.

Table 2: Contribution-evidence summary. The table states what each contribution is, which evidence supports it, and which stronger claim remains outside the current manuscript.

Contribution	Supporting evidence in this paper	Claim boundary
Identity-agenda confusion as a scholarly deduplication failure mode.	Problem formulation, hard-negative false-merge rate, and Open-v2 hard-negative evidence.	Identifies a concrete safety failure mode, not a prevalence estimate for all scholarly corpora.
IAD-Bench as a provenance-aware pair contract.	Gold, proxy, silver, and manual-validation layers with label source, label strength, provenance, split, and hard-negative fields.	Defines an evaluation contract; proxy and silver labels are not human identity gold.
IAD-Risk as a risk-aware merge mechanism.	Open-v2 evidence snapshot with single-space baseline HNFMR 0.790–0.999 and held-out IAD-Risk same-work F1=0.980, HNFMR=0.000.	Supports targeted false-merge suppression, not a broad method-ranking claim.

2. Related Work

Entity resolution research defines the broader pipeline in which scholarly deduplication is situated. Classical record linkage formalizes the decision of whether two records refer to the same entity [3], while modern surveys describe end-to-end entity resolution as a workflow involving blocking, matching, and clustering under scale and heterogeneity constraints [6]. System work such as Magellan further emphasizes that practical entity matching requires sampling, labeling, debugging, workflow management, and production execution rather than only a matching function [7]. IAD-Risk follows this pipeline view, but focuses on a specific safety gap in scholarly data: a pair

can be highly related in topic or citation context while still being an unsafe merge.

Neural entity matching methods provide the closest algorithmic baselines for learning pair-level match decisions. DeepER represents tuples with neural distributed representations and supports blocking over learned tuple vectors [8]. DeepMatcher explores design choices for neural entity matching models [4], and Ditto shows that transformer-based matching can benefit from pretraining, fine-tuning, and data augmentation [5]. RoBERTa-style pair encoders [9] are therefore natural baselines for this paper. The distinction is that these methods usually optimize a same-entity classification objective, whereas IAD-Risk adds explicit same-agenda and agenda-non-identity heads so that semantic relatedness can become blocking evidence rather than merge evidence.

Scientific document representation methods are strong baselines because they intentionally capture citation, topical, and semantic relatedness. SPECTER uses citation-informed supervision to learn document-level scientific representations and introduced SciDocs as a document-level evaluation benchmark [1]. SciNCL improves scientific representations with neighborhood contrastive learning [2], and SciRepEval broadens scientific representation evaluation across classification, regression, ranking, and search while releasing the SPECTER2 model family [10]. These methods are useful for candidate generation and relatedness scoring, but their optimization target is not conservative work identity. IAD-Risk therefore treats high scientific relatedness as a possible false-merge risk unless identity evidence is also strong.

Scientific language resources and open scholarly metadata make reproducible benchmark construction possible, but they do not remove the need for provenance-aware evaluation. SciBERT shows that domain-specific pre-training on scientific text improves scientific NLP tasks [11], and S2ORC provides metadata, abstracts, references, and structured full text for many open-access papers [12]. OpenAlex provides a large open index of scholarly works and concepts [13], while OpenCitations provides open citation data [14]. IAD-Bench uses these resources to construct same-work, same-agenda, and agenda-non-identity evidence, but separates gold, proxy, and silver labels so that citation or topic proximity is never presented as human identity gold.

Table 3: Positioning against the closest lines of work. IAD-Risk differs by treating agenda relatedness as a risk signal for merge decisions rather than as direct identity evidence.

Line of work	Primary optimization target	Limitation for scholarly deduplication	IAD-Risk distinction
End-to-end entity resolution systems	Blocking, matching, labeling, and deployment workflows.	Workflow abstractions do not separate topical relatedness from work identity.	Adds identity-agenda semantics and false-merge risk gates inside matching.
Neural entity matching	Pair-level same-entity classification from tuple or text encoders.	Strong text similarity can still link different papers in one agenda.	Trains same-work, same-agenda, and agenda-non-identity heads with provenance-aware masking.
Scientific document representations	Citation-informed or contrastive scientific relatedness embeddings.	Relatedness embeddings support retrieval but are unsafe as direct merge evidence.	Uses representation similarity as candidate and agenda evidence, then blocks risky merges.
Open scholarly metadata benchmarks	Public metadata, citation, and topic signals for scalable evaluation.	Topic or citation proximity is silver evidence, not human identity gold.	Reports gold, proxy, and silver strata separately and ties claims to label strength.

This positioning narrows the novelty claim. IAD-Risk is not a replacement for end-to-end entity resolution workflows; it inserts a risk-aware relation layer into scholarly matching when agenda relatedness can make automatic merges unsafe. It is also not a comparative ranking over all neural matching methods; the reported comparisons test the merge-safety framing under the available Open-v2 evidence snapshot. Finally, IAD-Bench does not claim that OpenAlex/OpenCitations silver evidence is human gold, but uses those rows as provenance-marked stress evidence for agenda-level false-merge risk.

3. Problem Formulation

Let d_i and d_j be two scholarly records. The target identity label y_{ij}^I indicates whether the records describe the same scholarly work. The agenda label y_{ij}^A indicates whether the records discuss a related research agenda.

A hard negative is a pair with high agenda evidence but negative identity evidence:

$$y_{ij}^A = 1, \quad y_{ij}^I = 0.$$

This pair should not be merged even if the documents are semantically close.

We evaluate merge safety with hard-negative false-merge rate:

$$\text{HNFMR} = \frac{\#\{(i, j) : \hat{m}_{ij} = 1, y_{ij}^A = 1, y_{ij}^I = 0\}}{\#\{(i, j) : y_{ij}^A = 1, y_{ij}^I = 0\}},$$

where $\hat{m}_{ij}$ is the automatic merge decision. The goal is not only to maximize same-work F1, but also to control HNFMR under explicit evidence provenance.

3.1. Notation and Relation Semantics

Table 4 summarizes the notation used throughout the paper. The key distinction is that identity variables describe whether two records denote the same work, while agenda variables describe topical or citation-level relatedness. This distinction prevents a high relatedness score from being interpreted as direct merge evidence.

Table 4: Notation and relation semantics. The paper uses separate symbols for identity, agenda relatedness, agenda-non-identity risk, and merge decisions.

Symbol or field	Meaning	Role in evaluation
d_i, d_j	Two scholarly records.	Input pair for scoring and merge decisions.
y_{ij}^I	Identity label: the records describe the same work.	Gold identity evidence when supported by the label source.
y_{ij}^A	Agenda label: the records share a research agenda.	Proxy or silver relatedness evidence, not sufficient for merging.
y_{ij}^{ANI}	Agenda-non-identity label.	Hard-negative supervision for pairs that are related but should not merge.
p_{work}	Predicted same-work probability.	Positive merge evidence.
p_{agenda}	Predicted agenda-relatedness probability.	Relatedness evidence that can increase risk when identity is weak.
p_{ani}	Predicted agenda-non-identity probability.	Direct risk evidence against automatic merging.
p_{risk}	Derived false-merge risk score.	Gate for rejecting or deferring risky pairs.
$\hat{m}_{ij}$	Automatic merge decision.	Determines FMR and HNFMR.
label strength	Gold, proxy, silver, or manual validation layer.	Controls how a row can be used in training, evaluation, and claims.

4. IAD-Bench

IAD-Bench is a benchmark contract for organizing scholarly pair evidence. It is not a claim that every label has the same evidential status. Each pair stores relation labels, expected identity labels, expected agenda labels, label source, label strength, provenance, split, and hard-negative level.

Table 5: IAD-Bench label layers. Gold, proxy, and silver labels are reported separately to avoid overstating evidence strength.

Layer	Source	Use in this paper
Gold	DeepMatcher / py_entitymatching	Same-work identity evaluation
Proxy	SciRepEval / SciDocs-style proximity	Same-agenda analysis
Silver	OpenAlex / OpenCitations	Agenda-non-identity hard negatives
Manual validation	Independent human review	External validation target

4.1. Document Schema Contract

IAD-Bench first standardizes each scholarly record before constructing pair evidence. Table 6 lists the document fields used by the manuscript. The contract keeps identity-bearing metadata, agenda-bearing metadata, and source provenance in one record so that downstream pair construction can be audited without redistributing raw third-party files.

Table 6: IAD-Bench document schema contract. Missing values are represented by empty strings, empty arrays, or null values rather than by dropping required fields.

Field group	Required fields	Interpretation role
Record identity	<code>document_id</code> , <code>source_dataset</code>	Makes each standardized record traceable to its originating public source.
Text and authorship	<code>title</code> , <code>abstract</code> , <code>authors</code>	Provides the textual and author evidence used by identity and agenda features.
Bibliographic metadata	<code>year</code> , <code>venue</code> , <code>doi</code> , <code>arxiv_id</code> , <code>openalex_work_id</code>	Supplies identifier and publication-context evidence for same-work decisions.
Agenda context	<code>topics</code> , <code>references</code>	Supplies topical and citation evidence used for same-agenda and hard-negative construction.

4.2. Pair Schema Contract

IAD-Bench uses a pair schema that makes relation semantics and evidence strength auditable. Table 7 lists the required fields used by the manuscript. The schema separates the binary same-work target from agenda relatedness, records the source and strength of each label, and keeps split and hard-negative metadata visible so that training, threshold selection, held-out reporting, and manual audit cannot be silently mixed.

Table 7: IAD-Bench pair schema contract. The schema records identity labels, agenda labels, provenance, split assignment, and hard-negative status for each pair.

Field group	Required fields	Interpretation role
Pair identity	<code>pair_id</code> , <code>source_document_id</code> , <code>target_document_id</code>	Makes every row traceable to two standardized documents and prevents anonymous aggregate-only reporting.
Relation targets	<code>relation_label</code> , <code>expected_label</code> , <code>expected_agenda_label</code>	Separates same-work identity from agenda relatedness before metrics are computed.
Evidence source	<code>label_source</code> , <code>label_strength</code> , <code>label_provenance</code>	Records whether a label is gold, proxy, silver, or human-audit evidence and what signals support it.
Evaluation control	<code>split</code> , <code>hard_negative_level</code>	Controls train/dev/test/audit use and identifies agenda-level hard negatives for HNFMR.

The Open-v2 benchmark variant used in the core evidence combines 415 DeepMatcher gold pairs with 10,000 OpenAlex-derived silver hard negatives, producing 10,415 pair records over 1,737 documents. The DeepMatcher component provides same-work gold labels, while the OpenAlex component stresses same-agenda non-identity behavior. This design is intentionally conservative: OpenAlex and OpenCitations are used to expose false-merge risk, not to replace human gold.

Table 8: Open-v2 benchmark composition used in the current evidence snapshot. The table separates identity evidence from hard-negative stress evidence before any metric is reported.

Evidence stratum	Pair count	Evaluation role	Claim boundary
DeepMatcher gold identity	415	Measures same-work matching ability.	Supports identity evaluation within the imported gold benchmark pairs.
OpenAlex and OpenCitations silver hard negatives	10,000	Stresses agenda-level false-merge behavior.	Supports hard-negative safety analysis, not human non-identity gold.
Open-v2 combined pair scope	10,415	Provides the shared pair contract for the reported evidence snapshot over 1,737 documents.	Supports the Open-v2 mechanism demonstration; broader source-heldout claims require additional artifacts.

5. Method

5.1. Overview

IAD-Risk converts scholarly deduplication into selective risk-aware matching. Given a candidate pair, the model estimates identity, agenda, and agenda-non-identity signals, then derives a false-merge risk score. The final merge decision is made only when identity evidence exceeds a same-work threshold and risk evidence remains below blocking thresholds. Figure 1 summarizes the workflow. The method section follows the same execution order: it first defines identity and agenda signals, then introduces the relation heads and risk score, then specifies the merge gate, algorithmic audit trail, operational boundary, and implementation details.

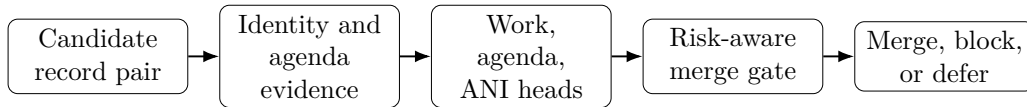


Figure 1: IAD-Risk pipeline. The framework first separates identity evidence from agenda evidence, then predicts relation signals and derives a false-merge risk score before applying a conservative merge gate.

5.2. Identity and Agenda Signals

IAD-Risk separates the representation space into identity-oriented and agenda-oriented evidence. Identity evidence includes strong identifiers, near-duplicate titles, author overlap, year and venue agreement, and work-level metadata. Agenda evidence includes shared research topics, similar abstracts, citation neighborhoods, shared references, and scientific representation similarity. The separation is necessary because the same feature can have different meanings: high embedding similarity may indicate same work, but it may also indicate two different papers in the same research area.

5.3. Multi-Task Relation Heads

The model predicts three pair-level relation quantities: p_{work} , p_{agenda} , and p_{ani} . The same-work head estimates identity. The same-agenda head estimates topical relatedness. The agenda-non-identity head detects pairs that are related but should not be merged. The false-merge risk score is then derived from these heads:

$$p_{\text{risk}} = \max\{p_{\text{ani}}, p_{\text{agenda}}(1 - p_{\text{work}})\}.$$

This construction makes the safety signal high when a pair is explicitly predicted as agenda-non-identity, or when it is agenda-related but lacks strong identity evidence.

5.4. Risk Score Design Rationale

p_{risk} is a conservative upper-envelope risk proxy for automatic merge decisions. The score increases monotonically with agenda-non-identity evidence through p_{ani} , and it also increases when agenda evidence is high and identity evidence is weak through $p_{\text{agenda}}(1 - p_{\text{work}})$. The max operator keeps either direct ANI evidence or indirect agenda-without-identity evidence sufficient to block automatic merging. This design matches the safety objective of the paper: a pair should defer rather than merge when either source of false-merge risk is high.

Table 9: Risk score design rationale. The score is designed as a conservative merge gate, not as an unconditional probability of error.

Design element	Design rationale	Boundary
p_{ani} term	Directly represents evidence that a related pair is not the same work.	Requires valid ANI supervision or artifact-backed hard-negative evidence.
$p_{\text{agenda}}(1 - p_{\text{work}})$ term	Flags pairs that are topically related but lack corresponding identity support.	It is not a calibrated probability unless validated against held-out artifacts.
Max operator	Makes either risk pathway sufficient to block a merge.	Conservative blocking can increase deferrals and must be tuned to the application.
Threshold gate	Converts the risk proxy into a merge, block, or defer decision.	Threshold transfer must be rechecked under new source distributions.

5.5. Training Objective

The learning objective combines identity, agenda, and agenda-non-identity supervision only through valid supervision channels. Let m_{ij}^w , m_{ij}^a , and m_{ij}^n indicate whether pair (i, j) has valid same-work, same-agenda, and agenda-non-identity supervision, respectively. The model minimizes the masked objective

$$\begin{aligned}
Z &= \sum_{(i,j)} (m_{ij}^w + m_{ij}^a + m_{ij}^n), \\
\mathcal{L} &= \frac{1}{Z} \sum_{(i,j)} [\lambda_w m_{ij}^w \text{CE}(y_{ij}^I, p_{\text{work}}) \\
&\quad + \lambda_a m_{ij}^a \text{CE}(y_{ij}^A, p_{\text{agenda}}) + \lambda_n m_{ij}^n \text{CE}(y_{ij}^{ANI}, p_{\text{ani}})],
\end{aligned}$$

where CE is binary cross-entropy and $\lambda_w, \lambda_a, \lambda_n$ control the contribution of each supervision channel. Missing labels therefore do not create negative examples for unrelated heads. A mini-batch with $Z = 0$ is skipped before the optimizer update, and each reported relation head requires positive mask coverage in the declared training split. DeepMatcher pairs supervise same-work identity, whereas OpenAlex/OpenCitations silver hard negatives supervise agenda-non-identity risk without being promoted to human gold identity labels. The false-merge risk score is not directly supervised; it is derived after the relation heads are estimated.

5.6. Risk-Aware Merge Decision

The merge policy is:

$$\hat{m}_{ij} = 1 \quad \text{iff} \quad p_{\text{work}} \geq \tau_w, \quad p_{\text{ani}} < \tau_a, \quad p_{\text{risk}} < \tau_r, \quad \text{cannot_link}_{ij} = 0.$$

This rule is deliberately selective. It can reject or defer high-risk pairs instead of forcing every semantically related pair into a binary match decision. The same decision interface also supports constrained union-find clustering, where cannot-link evidence can block documented conflicts when coverage is available. Operationally, the three outputs have different meanings. A pair is merged only when all gate conditions pass. A pair is blocked when explicit cannot-link evidence, high agenda-non-identity evidence, or high derived risk makes an automatic merge unsafe under the fixed operating point. A pair is deferred when identity evidence is insufficient for automatic merging but the available evidence does not justify a hard block. This distinction lets the method control false merges without claiming that every unresolved pair has been fully adjudicated.

5.7. Scoring and Merge Algorithm

The executable method follows a fixed scoring and merge order. For each candidate pair, IAD-Risk first builds identity, agenda, ANI, and audit fields from the pair schema. It then predicts the three relation heads, computes the derived risk score, applies cannot-link evidence, and emits one of three decisions: merge, block, or defer. Table 10 states the algorithmic contract used by the implementation and the artifact audit.

Table 10: Scoring and merge algorithm for IAD-Risk. The table gives the execution order that connects pair inputs, relation-head outputs, risk gating, and metric-counting artifacts.

Step	Operation	Required inputs or outputs	Audit role
1	Load a candidate pair and schema metadata.	Pair ID, document IDs, label strength, provenance, split, and hard-negative level.	Fixes row identity before scoring.
2	Build feature groups without using audit metadata as predictors.	Identity, agenda, ANI, and cannot-link feature groups.	Prevents provenance or split leakage.
3	Predict relation heads.	p_{work} , p_{agenda} , and p_{ani} .	Records interpretable relation evidence.
4	Compute derived false-merge risk.	$p_{\text{risk}} = \max\{p_{\text{ani}}, p_{\text{agenda}}(1 - p_{\text{work}})\}$.	Makes the blocking score reproducible.
5	Apply the merge gate and cannot-link evidence.	τ_w , τ_a , τ_r , cannot-link flag, and threshold source.	Emits merge, block, or defer under a fixed operating point.
6	Write prediction and metric audit fields.	Decision, row scope, denominators, thresholds, and checksum-bound artifact row.	Supports same-work F1, FMR, HNFMR, coverage, and defer-rate audits.

5.8. Operational Complexity and Net Benefit

IAD-Risk is appropriate when false merges are more costly than additional review. The additional cost comes from three relation heads and explicit threshold records, rather than from a new end-to-end clustering system. In an operational scholarly index, this cost is justified when a false merge can contaminate citation counts, literature-review evidence, or recommendation signals. In contrast, low-risk bulk cleanup tasks may prefer a simpler deduplication rule if occasional false merges have limited downstream effect.

Table 11: Operational complexity and net-benefit boundary. The framework is intended for safety-sensitive scholarly identity decisions, not as a universal replacement for simple deduplication pipelines.

Operational item	Required practice	Net-benefit boundary
Model complexity	Train and audit same-work, same-agenda, and ANI heads under one pair schema.	The net benefit is strongest in high-stakes scholarly indexes where false merges are costly.
Threshold governance	Select shared thresholds on validation evidence and record them in artifacts.	Thresholds are not tuned per pair or changed after test inspection.
Deferral policy	Record the deferral budget and manual-review capacity before deployment.	A high defer rate is acceptable only when review capacity exists.
Simple alternatives	Retain simple rules for low-risk bulk cleanup or identifier-only reconciliation.	IAD-Risk is not a universal replacement for simple deduplication pipelines.

5.9. Version and Identifier Boundary

Version and identifier evidence is treated as merge-policy input rather than as a post-hoc cleanup rule. DOI, arXiv, and OpenAlex identifiers provide strong identity cues when they agree, but publication-lineage evidence is still needed when records describe preprint, conference, and journal versions of related work. IAD-Risk therefore separates identifier agreement, identifier conflict, and uncertain version boundaries before applying cluster-level merging.

Table 12: Version and identifier boundary for merge decisions. The policy prevents related scholarly versions or conflicting identifiers from being silently collapsed into one work identity.

Evidence condition	Operational policy	Boundary statement
Identifier agreement	DOI, arXiv, or OpenAlex agreement is positive identity evidence when other metadata do not contradict it.	identifier agreement supports merge eligibility, but it does not replace source and metadata checks.
Identifier conflict	Conflicting DOI, arXiv, or OpenAlex fields are treated as risk evidence before any automatic merge.	identifier conflict creates cannot-link or defer evidence unless an audit log resolves the conflict.
Version relationship	Related preprint, conference, and journal versions require an explicit version policy.	not every related version is automatically the same work.
Ambiguous lineage	When lineage cannot be established from identifiers and metadata, the pair is not forced into a merge.	manual adjudication is required for ambiguous version boundaries.
Cluster propagation	Version decisions are recorded before transitive clustering.	version policy must be declared before cluster-level merging.

5.10. Failure-Control Rationale

IAD-Risk is designed as a failure-control framework rather than as another similarity scorer. Table 13 summarizes how each design choice addresses a specific false-merge pathway. The table also states the remaining boundary for each control, because a safety mechanism should make its unsupported cases explicit.

Table 13: Failure-control rationale of IAD-Risk. Each component targets a concrete way in which agenda-level similarity can be mistaken for work identity.

Failure pathway	Design response	Remaining boundary
Topically close papers receive high semantic similarity. Silver metadata is treated as if it were human gold.	Separate agenda evidence from identity evidence before the merge gate. Preserve label strength and provenance, and mask supervision by valid label source.	Requires the benchmark to include agenda-level hard negatives. Silver labels support stress testing, not exhaustive truth claims.
Pairwise errors contaminate clusters through transitivity.	Apply risk gating before constrained union-find and respect cannot-link evidence.	Cluster-level guarantees require complete cannot-link coverage.
Thresholds turn a classifier into an unsafe automatic merger.	Expose work, ANI, and risk thresholds and evaluate held-out merge decisions.	Threshold transfer should be rechecked under new source distributions.
Proxy labels are over-interpreted.	Keep proxy rows separate unless the label source supports the relation being evaluated.	Proxy rows remain non-human evidence even when reproducible.

5.11. Design Alternatives and Rejected Shortcuts

Several simpler designs are plausible but unsafe for the claim made in this paper. Table 14 explains why IAD-Risk uses explicit identity, agenda, and agenda-non-identity signals instead of treating scholarly deduplication as ordinary semantic thresholding or a single pair-classification problem. The purpose of this comparison is not to dismiss these alternatives; some are strong baselines in the reported evidence. The point is that they do not expose the relation semantics needed to audit agenda-driven false merges.

Table 14: Design alternatives considered during method design. Each rejected shortcut lacks an explicit mechanism for separating agenda relatedness from work identity or for auditing false-merge risk.

Alternative	Why it is insufficient for this failure mode	IAD-Risk design response	Evidence boundary
Tune a representation-similarity threshold.	High scientific relatedness can still connect different works in the same task, benchmark, or citation community.	Use representation similarity as candidate and agenda evidence, not direct merge evidence.	Supports false-merge-risk motivation, not a full retrieval benchmark.
Use one supervised pair classifier.	A strong classifier can reduce false merges but does not expose whether a positive score comes from identity or agenda evidence.	Predict same-work, same-agenda, and agenda-non-identity signals separately before gating.	RoBERTa remains a strong baseline; broad superiority is not claimed.
Use provenance as a model feature.	Source or split identifiers can create leakage and make held-out interpretation fragile.	Keep provenance as audit metadata and apply provenance-aware supervision masking.	Source-heldout claims require stronger source coverage.
Always force a binary merge decision.	Some pairs have related evidence but insufficient identity evidence, making automatic merging risky.	Emit merge, block, or defer through the risk-aware gate.	Defer behavior needs deployment-specific review policies.
Select thresholds after test results.	Post-hoc threshold choice can hide false-merge risk and inflate held-out conclusions.	Fix operating points before final reporting and require threshold artifacts for stronger claims.	Threshold stability needs a released grid and checksums.

5.12. Implementation Details

The reported transformer variants use frozen scientific document encoders rather than end-to-end fine-tuning. For each candidate pair, the implemen-

tation builds three feature groups: identity features combine transformer distances with identifiers, author overlap, venue, and year signals; agenda features combine semantic similarity with topic and citation overlap; agenda-non-identity features add conflict indicators such as different identifiers and year conflict. The three relation heads are trained on the training split, and the merge policy exposes τ_w , τ_a , and τ_r as configurable thresholds. Unless a validation protocol overrides them, the implementation uses 0.5 for all three thresholds.

5.13. Feature and Head Specification

The implementation uses label provenance for supervision control, not as a predictive shortcut. Pair metadata such as label source, label strength, provenance, split, and hard-negative level are copied to prediction outputs for auditing, but they are excluded from the identity, agenda, and risk feature groups used by the relation heads. Table 15 gives the implementation-level contract used by the reported transformer variants.

Table 15: Feature and head specification for IAD-Risk transformer variants. Metadata fields are retained for auditing but are not used as model features.

Head or stage	Main input fields	Supervision or calculation	Output role
Identity head	Transformer distances, title similarity, author overlap, first-author match, venue similarity, year match, DOI/arXiv/OpenAlex identifier agreement.	Same-work labels from valid identity evidence, with provenance-aware masking.	Estimates p_{work} as positive merge evidence.
Agenda head	Transformer cosine, title and venue similarity, OpenAlex-style topic overlap, reference Jaccard similarity, year-distance score.	Same-agenda proxy or silver evidence when the source supports agenda relatedness.	Estimates p_{agenda} as relatedness evidence, not direct merge evidence.
ANI risk head	Transformer distances, title and author signals, topic overlap, reference overlap, year conflict, identifier agreement, and different-identifier conflicts.	Agenda-non-identity hard negatives from valid hard-negative sources.	Estimates p_{ani} as direct blocking evidence.
Risk gate	p_{work} , p_{agenda} , p_{ani} , τ_w , τ_a , τ_r , and cannot-link flags.	Computes $p_{\text{risk}} = \max\{p_{\text{ani}}, p_{\text{agenda}}(1 - p_{\text{work}})\}$.	Emits merge, block, or defer decisions.
Audit metadata	Pair ID, source pair ID, label strength, label source, provenance, split, and hard-negative level.	Copied from pair records and result artifacts.	Supports artifact review; it is not a training feature.

6. Experiments

6.1. Dataset Construction and Evidence Strength

Open-v2 is built by converting public scholarly metadata into a unified pair schema. DeepMatcher/py_entitymatching records provide the same-work gold layer, while OpenAlex topics and OpenCitations-style citation signals provide agenda-level silver hard negatives. Each pair keeps its label strength, provenance, split, and hard-negative flag so that a gold identity

result is never averaged with a silver stress-test result without being identified as such.

The benchmark therefore has two roles. The gold layer measures ordinary duplicate matching ability, and the silver hard-negative layer measures whether a system over-merges topically close but distinct papers. This separation is important for interpreting the results: a representation baseline can look useful for scientific retrieval while still being unsafe for automatic work identity merging.

6.2. Split and Leakage Controls

The evaluation protocol treats split assignment as part of the evidence contract rather than as a preprocessing detail. Each pair carries a split field, label strength, label source, relation label, provenance, and hard-negative level. Training uses only the declared training split, threshold selection uses validation evidence when available, and held-out metrics are reported only after the operating point has been fixed. Metadata fields that identify source, provenance, or split are retained for auditing but are excluded from model features.

Table 16: Split and leakage controls used to interpret IAD-Bench results. The controls define what the current evidence supports and what stronger follow-up evidence would require.

Control	Manuscript role	Stronger evidence boundary
Train/dev/test split field	Separates fitting, threshold selection, and held-out reporting.	Full numerical reproduction requires released prediction files, configs, and split summaries.
Unordered pair leakage guard	Detects whether the same document pair appears across multiple splits.	A blocked leakage audit prevents held-out claims until duplicate pair assignments are removed.
Label-stratum coverage audit	Checks whether gold, proxy, and silver evidence appear in the intended evaluation strata.	Missing strata must be reported as evidence gaps rather than averaged into a single score.
Source-heldout readiness audit	Tests whether relation labels have enough independent label sources for source-heldout claims.	Broad source generalization remains conditional until source-heldout coverage is defensible.
Topic-heldout readiness audit	Tests whether enough topics exist to reserve unseen-topic evaluation slices.	Cross-topic stability should not be claimed when topic coverage is insufficient.

These controls reduce two common validity risks. First, the model cannot gain accuracy from explicit provenance or split identifiers because those fields are audit metadata rather than predictive features. Second, a held-out result is interpreted only within the split and coverage regime that generated it. If a split-readiness audit blocks source-heldout or topic-heldout claims, the manuscript keeps the conclusion at the level of random-split or mechanism evidence instead of presenting it as broad generalization.

6.3. Experimental Questions

The experiments ask four questions. RQ1 tests whether IAD-Risk preserves same-work matching performance on gold labels. RQ2 tests whether it reduces false merges on agenda-level hard negatives. RQ3 examines whether the observed behavior is consistent with the proposed risk mechanism. RQ4 tests whether results remain interpretable under gold, proxy, and silver label strata.

Table 17: Evaluation protocol. Each question is tied to a label stratum and a metric so that evidence strength is not mixed across sources.

RQ	Evidence layer	Metric	Interpretation
RQ1	Gold identity pairs	Same-work F1	Duplicate matching ability
RQ2	Silver hard negatives	HNFM	False-merge safety
RQ3	Mechanism analysis	FMR and HNFM	Role of risk signals
RQ4	Gold/proxy/silver strata	Split metrics	Provenance-aware robustness

6.4. Evaluation Metrics

Same-work F1 is computed on pairs whose identity labels come from the gold layer. Ordinary false-merge rate (FMR) is the fraction of non-identity pairs that the system marks for automatic merging. Hard-negative false-merge rate (HNFM) is the same error measured only on agenda-level hard negatives, where topic or citation evidence is high but identity evidence is negative. Reporting both FMR and HNFM separates general matching errors from the specific identity-agenda confusion that this paper studies.

6.5. Decision-to-Metric Mapping

The reported metrics treat the selective merge decision as a pair-level prediction. In this mapping, automatic merge is the positive decision, while block and defer are non-merge decisions. Deferred same-work pairs reduce recall in same-work F1, so a system cannot improve F1 by deferring positive identity pairs without penalty. FMR and HNFM count only automatic merges among non-identity rows, while coverage and defer rate must be reported separately before operational automation claims are made.

Table 18: Decision-to-metric mapping for selective merge outputs. The mapping prevents block and defer decisions from being hidden inside false-merge metrics.

Decision output	Metric treatment	Interpretation boundary
Merge	Positive prediction for same-work F1; counted as an error for FMR/HNFMR when the label is non-identity.	Measures automatic merge behavior under the fixed operating point.
Block	Non-merge prediction.	Suppresses false merges but does not count as resolving a positive same-work pair.
Defer	Non-merge prediction.	Protects against unsafe automatic decisions but shifts the case to review and lowers recall when the pair is truly same-work.

6.6. Metric Formula Boundary

The metric formulas use the selective merge output after thresholding. Let TP denote same-work pairs that are automatically merged, FP denote non-identity pairs that are automatically merged, and FN denote same-work pairs that are blocked or deferred. Same-work F1 is computed as $2TP/(2TP + FP + FN)$ on the gold identity evaluation rows. FMR denominator is all non-identity rows in the evaluated scope, while HNFMR denominator is the agenda-level hard-negative subset. Rows excluded by missing labels are not silently added to denominators; they remain outside the corresponding metric until a valid label stratum is available.

Table 19: Metric formula boundary. Each metric is tied to a denominator and a label stratum so that missing labels, deferrals, and hard-negative rows are not mixed silently.

Metric	Formula or denominator	Boundary
Same-work F1	$2TP/(2TP + FP + FN)$ over gold identity rows.	Defer and block decisions on true same-work rows enter FN.
FMR	Automatic merges among all non-identity rows divided by all non-identity rows in the evaluated scope.	Measures unsafe automatic merges, not manual-review workload.
HNFMFMR	Automatic merges among agenda-level hard negatives divided by the agenda-level hard-negative subset.	Measures identity-agenda false merges under the hard-negative stratum.

6.7. Threshold Selection and Uncertainty Reporting

Threshold-sensitive systems are evaluated under a fixed selection protocol rather than by choosing the best test threshold after observing results. Merge thresholds are selected on validation evidence when validation rows are available; otherwise, the default implementation threshold is reported as a fixed operating point. The test split is then used only for final metric reporting. This protocol is necessary because a small threshold change can alter FMR and HNFMFMR more strongly than it alters same-work F1.

Table 20: Threshold and uncertainty reporting protocol. The protocol separates what is required for the current main table from what is required for stronger follow-up claims.

Item	Reporting purpose	Interpretation role
Merge thresholds	Record τ_w , τ_a , τ_r , validation source, and fixed operating point.	Prevents post-hoc threshold selection on the test set.
Metric uncertainty	Report bootstrap intervals when prediction files and resampling logs are present in the artifact package.	Distinguishes stable effects from small numerical differences.
Risk metrics	Always report FMR and HNFMR with same-work F1.	Prevents a high F1 score from hiding unsafe hard-negative merges.
Ablation claims	Claim component causality only when no-risk-gate, no-ANI, and single-space variants are present.	Separates mechanism-consistent evidence from complete causal ablation evidence.
Artifact audit	Tie each numerical table to predictions, configs, logs, checksums, and commit identifiers.	Makes result-level reproduction possible without committing raw data.

6.8. Statistical Interpretation Boundary

The reported Open-v2 numbers are point estimates for a fixed evidence snapshot, not statistical superiority estimates. Confidence intervals, significance tests, and model-ranking statements are intentionally withheld unless the released artifact package contains the exact prediction files, resampling logs, random seeds, and checksums used to compute them. In particular, an HNFMR value of 0.000 means that no hard-negative false merge was observed under the reported pair scope and operating point; it should not be read as proof of zero risk under all source distributions.

Table 21: Statistical interpretation boundary for the Open-v2 evidence snapshot. The table states how reported numbers may be read before interval and significance artifacts are released.

Interpretation item	Current manuscript status	Required evidence before stronger claim
Point estimates	Reported for the fixed Open-v2 scope, split, and operating points.	Prediction files, metric summaries, thresholds, and checksums for the same rows.
Confidence intervals	Not reported as primary evidence in the current manuscript.	bootstrap_intervals generated from the same prediction files, with command logs and random seeds.
Statistical significance	Not claimed for baseline ranking or model superiority.	Predefined tests, multiplicity handling, input artifacts, and reproducible analysis logs.
Zero HNFMR rows	Interpreted as no observed hard-negative false merge in the reported scope.	Larger source-heldout evidence and manual validation before any zero-risk wording.
Model ranking	Restricted to mechanism interpretation rather than a universal ranking.	Same-scope predictions, interval estimates, ablations, and manual-validation slice.

6.9. Operating Point Disclosure

The main result table reports fixed operating points rather than post-hoc best test thresholds. Table 22 summarizes how each row family obtains its decision field and operating point. Exact threshold values, selected validation sources, and command logs must be verified from the released artifact manifest before using the table for numerical reproduction.

Table 22: Operating point disclosure for the Open-v2 result table. The table records how merge decisions are produced and what artifact is required for threshold audit.

Row family	Decision field	Operating point source	Audit requirement
Representation cosine baselines	Cosine score threshold on scientific document embeddings.	Fixed baseline metric summary; threshold chosen before final test reporting.	Score file, metric summary, and threshold entry.
RoBERTa pair classifier	Pair match probability threshold.	Fixed pair-classifier metric summary under the same pair schema.	Prediction file, metric summary, and model log.
IAD-Risk transformer variants	p_{work} , p_{ani} , and p_{risk} under the risk gate.	Model configuration; default $\tau_w = \tau_a = \tau_r = 0.5$ unless the artifact config records an override.	Prediction file, model JSON, thresholds, and checksums.

6.10. Selective Decision Coverage Boundary

IAD-Risk is a selective decision system, so false-merge metrics should be interpreted together with coverage and review-load evidence. A low FMR or HNFMR is not by itself a throughput claim if the operating point blocks or defers many pairs. Table 23 defines the additional quantities that must be released before making operational automation claims. The current manuscript therefore reports false-merge control under fixed operating points, but does not claim throughput reduction and does not claim all-pair automatic resolution.

Table 23: Selective decision coverage boundary. Coverage and review-load quantities must be computed from the same prediction files as the reported risk metrics before operational deployment claims are made.

Quantity	Required artifact source	Interpretation boundary
Automatic merge coverage	Merge decisions from the same prediction files and thresholds.	Needed before claiming that the system resolves a large fraction of pairs automatically.
Block rate	Blocked decisions under the reported risk gate.	Separates safety-driven blocking from improved identity recognition.
Defer rate	Deferred decisions under the fixed operating point.	High defer rate shifts work to manual review and must be reported with FMR/HNFMR.
Review load	Counts by label stratum, source, and hard-negative level.	Needed before claiming operational throughput reduction or review-cost savings.

6.11. *Pair-to-Cluster Evidence Boundary*

The reported pair-level metrics do not by themselves prove cluster-level deduplication quality. Pair decisions can propagate through transitive merge propagation when a deduplication system turns accepted links into connected components. A single unsafe merge can contaminate a larger cluster, while one cannot-link decision can block a transitive path that would otherwise merge distinct works. Table 24 states the additional audit evidence required before pair-level false-merge control can be interpreted as a cluster-level deployment guarantee.

Table 24: Pair-to-cluster evidence boundary. Cluster-level deployment claims require artifacts beyond pair-level prediction and metric files.

Evidence item	Required artifact source	Interpretation boundary
cluster assignments	Released component or cluster ID for each evaluated record.	Needed before computing cluster purity, split errors, or merged-work contamination.
cannot-link violations	<code>cannot_link_audit</code> generated from the same merge decisions and identifier-conflict rules.	Needed before claiming that risk gates prevent transitive false merges.
Cluster metric summary	<code>cluster_metric_summary</code> with cluster contamination rate, singleton rate, merge coverage, and conflict counts.	Needed before claiming cluster-level quality rather than pair-level risk control.
Pair-to-cluster trace	Mapping from pair decisions, thresholds, and cannot-link flags to final components.	Required to audit whether a pair-level error changes a downstream cluster.
Current manuscript claim	Pair-level F1, FMR, and HNFMR under fixed operating points.	The paper does not claim cluster-level contamination is eliminated.

6.12. Threshold Sensitivity Evidence Status

Threshold stability is treated as an audit requirement, not as an unsupported robustness claim. The current Open-v2 table reports fixed operating points and therefore supports the mechanism interpretation under those operating points. A stronger statement that the method is stable across thresholds would require a threshold grid computed from the same prediction files, predefined threshold ranges, metric summaries for each setting, and checksums in the released artifact manifest.

Table 25: Threshold sensitivity evidence status. The table separates the current fixed-threshold claim from the additional artifact evidence required for threshold-stability claims.

Audit item	Current manuscript status	Required artifact before stronger claim
Fixed operating point	Reported through the operating-point disclosure table and tied to validation or default thresholds.	Configuration file recording τ_w , τ_a , τ_r , baseline thresholds, and selection source.
Threshold grid	The threshold grid is not reported as primary evidence in the current manuscript.	Sensitivity table generated from the same prediction files over predefined threshold ranges.
Metric stability	Interpreted only for the reported operating points.	Per-threshold F1, FMR, HNFMR, pair counts, random seeds, and command logs.
Artifact manifest	Required for numerical audit but not included in Git.	Manifest and checksums that bind prediction files, summaries, thresholds, and code commit.
Interpretation boundary	Supports fixed-threshold false-merge control, not threshold-stable ranking across all operating points.	Stronger threshold-stability wording only after the released artifact contains the grid evidence.

6.13. Baselines

The main table reports two scientific representation cosine baselines and one supervised pair-classifier baseline: SciNCL cosine, SPECTER2 adapter cosine, and RoBERTa pair classification. These rows are evaluated through the same pair schema so that same-work F1 and hard-negative false-merge rate can be interpreted without mixing label strengths. Other repository baseline utilities are not used as primary manuscript evidence unless their metric summaries, prediction files, thresholds, and checksums are included in the released artifact package.

6.14. Baseline Inclusion Rationale

The baseline set is selected by evidence availability and evaluation relevance rather than by convenience. Table 26 states which baseline families are used as primary evidence and which are kept outside the main result

table until their artifacts are complete. This prevents utility code, fixture-level checks, or partially audited runs from being presented as if they were complete comparison experiments.

Table 26: Baseline inclusion rationale. A baseline enters the primary result table only when it has auditable outputs under the reported pair scope.

Baseline family	Current role	Inclusion or exclusion rule
Exact identifier matching	Important sanity-check family for DOI, arXiv, and publication-lineage evidence.	Excluded from primary result table when only utility code or fixture-level checks are available; full metrics require prediction files and checksums.
Title-normalization rules	Useful lower-bound family for metadata-only deduplication.	Excluded until the same pair scope, threshold records, metric summaries, and error cases are released.
Traditional entity-resolution systems	Relevant workflow baselines for blocking, matching, and clustering comparisons.	Included as primary evidence only when metric summaries, prediction files, threshold records, and checksums are available.
Scientific representation baselines	Primary stress-test evidence for single-space scientific relatedness.	Included because SciNCL cosine and SPECTER2 adapter cosine have reported score summaries under Open-v2.
RoBERTa pair classification	Strong supervised comparator for pair-level matching.	Included as a primary supervised comparator while keeping the result below broad ranking claims.

Omitting a baseline family from Table 31 is not a claim that omitted baselines were outperformed. It means the current manuscript does not yet have the artifact-backed evidence needed to make that row auditable. A same-scope baseline matrix is required before broad ranking claims, especially for exact identifier matching, title-normalization rules, and traditional entity-resolution systems.

6.15. Baseline Fairness Controls

Baseline comparison is controlled through the pair contract rather than through a broad comparative-ranking claim. All baselines consume the same IAD-Bench pair records for the row scope being reported, and each row uses

the same train/dev/test split field when training is required. Threshold-sensitive rows use validation-selected operating points before held-out reporting. All systems report same-work F1, FMR, and HNFMR from the same metric implementation.

Table 27: Baseline fairness controls for the Open-v2 evidence snapshot. The controls make the reported comparison auditable while preserving the scope boundary of each row family.

Control	Application in the reported table	Interpretation boundary
Input rows	Baselines use the same IAD-Bench pair records within each reported row scope.	A stricter ranking requires same-scope released prediction files for all rows.
Split use	Trainable rows use the same train/dev/test split field; non-trainable representation rows are reported as fixed score baselines.	Different available scopes should not be read as a single comparative ranking.
Threshold selection	Threshold-sensitive rows use validation-selected operating points or recorded fixed thresholds.	Post-hoc test-threshold selection is not used for the reported claims.
Feature boundary	Label source, provenance, and split identifiers are audit fields, not predictive features.	Provenance can explain evidence strength but cannot be used as a shortcut feature.
Metric implementation	All systems report same-work F1, FMR, and HNFMR from the same metric implementation.	Metric comparability does not remove the need for released predictions and checksums.

6.16. Implementation and Reproducibility

All reported systems are evaluated through the same pair-level schema, split field, and metric implementation. Threshold-sensitive baselines select merge thresholds on validation evidence and are then evaluated on held-out evidence. The main metrics are same-work F1 on gold identity pairs, ordinary false-merge rate, and HNFMR on agenda-level hard negatives. Rows with different pair counts in Table 31 should therefore be read as evidence for the same mechanism under different available result scopes, not as a single comparative ranking.

The repository separates executable code from large data artifacts: raw third-party data remain outside Git, while processing scripts, fixture tests, schema contracts, and artifact-release rules remain version controlled. Reproduction starts from public sources, rebuilds IAD-Bench pair files through the documented CLI, and checks generated outputs with manifests and checksums. This organization is intended to make the main evidence reproducible without redistributing restricted or large source files. The supplementary material states the claim-evidence boundary for each major conclusion so that code-level reproducibility, artifact-level reproducibility, and manual-validation requirements are not conflated.

6.17. Artifact Reproduction Protocol

Because the repository does not redistribute large third-party data, reproducibility is divided into code-level reproduction and result-level artifact reproduction. Table 28 defines what each level proves. The key principle is that small fixtures verify the processing contracts, while the full reported tables require either independently obtained public source files or a released artifact package with manifests and checksums.

Table 28: Reproduction levels for auditing the repository and the reported evidence.

Level	Required input	Audit target
L0 code check	Source tree, package environment, and manuscript scripts	Verify that the CLI, tests, public-release scan, and LaTeX build run in a clean environment.
L1 fixture rebuild	Public miniature fixtures under <code>tests/fixtures</code>	Verify that DeepMatcher, SciRepEval-style, OpenAlex-style, and IAD-Bench adapters produce the documented pair schema.
L2 public-source rebuild	Independently obtained public raw files from the cited sources	Rebuild derived IAD-Bench files with the documented data-processing commands and inspect label provenance.
L3 result audit	Released tables, predictions, logs, manifests, and checksums	Verify the numerical results reported in the manuscript against immutable artifact files.

6.18. Claim-Evidence Boundary for Result Interpretation

The result interpretation follows a conservative claim-evidence boundary. Each reported conclusion must be tied to a label stratum, an output artifact, and a statement of what the evidence does not prove. This boundary is especially important because the benchmark combines gold identity pairs with proxy and silver stress-test evidence rather than treating all rows as one homogeneous test set.

Table 29: Claim-evidence boundary used to interpret the reported results. Each claim class is paired with the supporting evidence and the claims that are intentionally not made.

Claim class	Required support	Boundary not claimed
Identity-agenda confusion is a concrete false-merge pathway.	Hard-negative false-merge rate on agenda-level silver pairs, compared with single-space scientific representation baselines.	A prevalence estimate for all scholarly corpora.
IAD-Risk reduces risky automatic merges in the reported setting.	Held-out Open-v2 IAD-Risk predictions, thresholds, metric summaries, and checksums.	A broad method-ranking claim or a claim that every source distribution is covered.
IAD-Bench supports provenance-aware evaluation.	Pair schema fields for relation type, label source, label strength, provenance, split, and hard-negative level.	A claim that proxy or silver rows are human-gold labels.
Repository-level reproduction is possible without committing raw data.	Public code, fixture tests, documented data-processing commands, manifest rules, and PDF build scripts.	Full numerical reproduction without public source files or a released artifact package.

6.19. Result Audit Trail

Table 31 should be read together with the artifact boundary in Table 29. Each numerical row must be traceable to three audit objects: a prediction

or score file, a metric summary produced from that file, and a checksum or manifest entry that fixes the artifact version. The same audit record must also contain per-row denominator counts for same-work F1, FMR, and HNFMR, the per-row threshold source, the scope label used in the main table, and the selective-decision coverage fields: automatic merge count, block count, defer count, automatic merge coverage, and defer rate. Prediction JSONL artifacts must expose system, pair/document IDs, expected labels, label strength, hard-negative level, split identifiers, score or probability fields, threshold source, threshold values where applicable, and merge decisions; threshold logs must expose threshold name, value, selection split, selection metric, selection rule, applied scope, and score field. The Git repository verifies the schema, fixture rebuild, public-release boundary, manuscript build, and submission package, but it deliberately does not store raw third-party data, large model outputs, or full prediction files. Therefore the manuscript treats the Open-v2 table as a supported evidence snapshot and reserves full numerical reproduction for the released L3 artifact package.

Table 30: Result artifact crosswalk for the Open-v2 evidence snapshot. Each row family in the main table has a required artifact path or manifest entry before it can be used for full numerical audit.

Main-table evidence	Required artifact IDs	Audit purpose
Open-v2 aggregate table	<code>open_v2_main_results</code> , <code>iad_bench_split_summary</code>	Verifies row scopes, pair counts, document counts, label strata, metric columns, and selective-decision coverage fields.
Representation cosine baselines	<code>representation_baseline_scores</code> , <code>threshold_selection_logs</code>	Verifies SciNCL and SPECTER2 scores, threshold choices, and full-pair false-merge rates.
RoBERTa pair classifier	<code>supervised_baseline_predictions</code> , <code>threshold_selection_logs</code>	Verifies supervised baseline predictions under the same pair schema and fixed operating point.
IAD-Risk transformer variants	<code>iad_risk_predictions</code> , <code>threshold_selection_logs</code>	Verifies held-out predictions, relation-head scores, risk-gate decisions, thresholds, and split identifiers.
Optional stronger claims	<code>bootstrap_intervals</code> , <code>ablation_suite</code> , <code>manual_validation_slice</code>	Required only before claiming confidence intervals, component causality, or human label precision.
Threshold-stability claims	<code>threshold_sensitivity_grid</code> , <code>threshold_selection_logs</code>	Required only before claiming stability across operating points or threshold ranges.

6.20. Main Open-v2 Results

Table 31: Open-v2 evidence snapshot. Lower false-merge rate is better. Rows are reported only when supported by current output summaries; the scope type column marks whether a row uses the full available Open-v2 scope or the held-out Open-v2 test scope. The Denom. audit column means that same-work F1, FMR, and HNFMR denominators must be present in the corresponding `open_v2_main_results` artifact row before numerical audit.

System	Scope type	Pairs Denom. audit	F1 ↑	FMR ↓	HNFMR ↓
SciNCL cosine	Full Open-v2	10415 Artifact row	0.054	0.785	0.790
SPECTER2	Full Open-v2	10415 Artifact row	0.044	0.999	0.999
adapter cosine	Full Open-v2	10415 Artifact row	0.825	0.001	0.0001
RoBERTa pair classifier	Held-out test	1042 Artifact row	0.980	0.001	0.000
IAD-Risk (SciNCL)	Held-out test	1042 Artifact row	0.980	0.001	0.000
IAD-Risk (SPECTER2)	Held-out test	1042 Artifact row	0.980	0.001	0.000

The Open-v2 evidence shows why identity-agenda disentanglement is necessary. Single-space scientific representation baselines preserve topical relatedness, but their merge threshold can produce high false-merge rates on OpenAlex silver hard negatives. IAD-Risk transformer variants keep high same-work F1 on the held-out Open-v2 test split while suppressing hard-negative false merges. RoBERTa pair classification is a strong baseline and narrows the gap substantially; therefore the paper does not make broad superiority claims. Instead, the main result is that relation heads plus an explicit risk score provide a transparent and controllable mechanism for false-merge suppression.

6.21. Result Interpretation Guardrails

Table 32 states how the Open-v2 table should be read before external artifact release and same-scope prediction files are available. The main result table includes Scope type labels to distinguish full available Open-v2 scope rows from held-out Open-v2 test scope rows. Scope labels prevent ranking interpretation when row families are supported by different available prediction scopes. The intended interpretation is mechanism-oriented; the table

labels this as a Mechanism-supported reading rather than a ranking-oriented reading.

Table 32: Result interpretation guardrails for the Open-v2 evidence snapshot. The table separates direct evidence, mechanism-supported reading, and unsupported reading.

Result row family	Directly supported reading	Unsupported reading
Representation baselines	The representation rows test false-merge exposure when one-space scientific relatedness is used as merge evidence.	They do not define the best possible representation-threshold system under a fully tuned, same-scope comparative ranking.
RoBERTa pair classifier	The RoBERTa row is a strong supervised comparator that shows pair classification can suppress many hard-negative merges.	It is not a component-level explanation of identity-agenda risk control.
IAD-Risk transformer variants	The IAD-Risk rows test split-held-out risk gating with explicit identity, agenda, and ANI signals.	The rows are not a claim of broad method superiority without same-scope predictions, intervals, and artifact checksums.
Whole table	The table gives a mechanism-supported reading that semantic relatedness alone can be unsafe and risk gating can block hard-negative merges in the reported setting.	The table is not a same-scope comparative ranking and is not evidence of threshold stability or zero risk.

6.22. Manual Validation Boundary

Manual validation is not completed in the current manuscript package. Silver hard negatives are stress-test evidence for identity-agenda false-merge risk, but they are not human-gold non-identity labels. Table 33 states what the current evidence supports and what must be added before making stronger label-precision claims.

Table 33: Manual validation boundary for interpreting silver hard negatives. The current manuscript reports stress-test evidence and reserves human-validation claims for a released reviewed slice.

Evidence item	Current role	Requirement before stronger claim
Silver hard negatives	Stress-test evidence for agenda-related false-merge risk.	A 500–1,000 pair reviewed slice covering hard negatives, high-score false-merge candidates, blocked or deferred pairs, version boundaries, identifier conflicts, and sparse metadata.
Human review process	Specified as a future validation layer rather than current result evidence.	The protocol requires two independent reviewers to inspect records while blind to model scores and merge decisions.
Adjudication artifacts	Not present in the current Git package or main result table.	The release must include an adjudication log, an agreement report, pair-level notes, and checksums before claiming label precision.
Manuscript claim	Supports bounded false-merge stress testing under Open-v2.	The paper does not claim complete human validation or that silver rows replace human-gold non-identity labels.

6.23. Scope Compatibility of the Open-v2 Table

The Open-v2 table is a scope-bounded evidence table, not a single comparative ranking. Full pair-scope representation baselines test whether single-space scientific relatedness can over-merge agenda-level hard negatives across the available Open-v2 pairs. Held-out IAD-Risk rows test whether risk-gated transformer variants preserve same-work performance and suppress hard-negative merges under a split-based model-evaluation protocol. The valid comparison is therefore mechanistic: semantic relatedness alone is unsafe as automatic merge evidence, while explicit risk gating can block hard-negative merges in the reported held-out setting.

Table 34: Scope compatibility for interpreting the Open-v2 evidence snapshot. The table states what each row family supports and which stronger comparison remains outside the current evidence.

Row family	Pair scope	Supported interpretation	Not supported
Representation baselines	Full available Open-v2 pair scope.	Tests whether one-space scientific relatedness creates hard-negative merge risk.	A tuned neural pair-classifier ranking under the same held-out prediction file.
RoBERTa pair classifier	Full available Open-v2 pair scope.	Shows that a strong supervised pair classifier can suppress most hard-negative merges.	A component-level explanation of identity-agenda risk control.
IAD-Risk transformer variants	Held-out Open-v2 test scope.	Tests whether explicit identity, agenda, and ANI heads support risk-gated held-out decisions.	A general ranking over every baseline before all rows share the same released prediction scope.
Future stronger comparison	Same released prediction scope plus manual-validation slice.	Would support a stricter comparative ranking and stronger label-precision claims.	A claim that this stronger comparison has already been completed by the current evidence snapshot.

This scope boundary is important for interpretation. The evidence is sufficient for the paper’s conservative claim that identity-agenda disentanglement is a practical false-merge control mechanism. It is not sufficient for a broad ranking claim until all row families are regenerated with the same released prediction scope, threshold-selection logs, checksums, and an independent manual-validation slice.

6.24. Extended Protocols and Boundaries

Open-v3 and source-heldout protocols are retained as follow-up evaluation paths, not as additional result evidence in the current manuscript package. These protocols are useful because they specify how source-diverse and domain-diverse validation should be audited before broader claims are made.

The manuscript therefore treats Open-v2 as the core mechanism demonstration and reserves Open-v3/source-heldout conclusions for a released artifact package that contains matched prediction scopes, threshold logs, checksums, and manual-validation evidence.

7. Mechanism and Error Analysis

The central mechanism question is whether false-merge control follows from explicit agenda-non-identity supervision and the derived risk score. The supported Open-v2 comparison gives mechanism-consistent evidence: single-space scientific representations can over-merge agenda-related hard negatives, while IAD-Risk makes agenda-non-identity evidence a direct blocking signal. This does not by itself establish a complete causal ablation. A complete ablation package must compare the full gate with variants that remove risk gating, remove agenda-non-identity supervision, or collapse identity and agenda evidence into one score.

Table 35: Mechanism evidence and interpretation boundary. The table separates current evidence from ablation outputs that would be required for a stronger causal claim.

Mechanism question	Current evidence	Interpretation boundary
Does topical relatedness create merge risk?	Representation baselines show high false-merge rates on OpenAlex silver hard negatives.	Supports a targeted failure mode, not a universal prevalence estimate.
Does explicit risk gating suppress hard-negative merges?	IAD-Risk transformer variants keep low HNFMR under the reported Open-v2 split.	Supports the reported contract and thresholds; transfer must be rechecked on new source mixes.
Is the gain caused by each IAD-Risk component?	The current manuscript reports mechanism-consistent HNFMR evidence.	A full causal claim requires ablations removing risk gating, ANI supervision, and single-space separation.
Can cluster-level contamination be eliminated?	The method supports cannot-link-aware constrained union-find.	Guarantees require sufficient cannot-link coverage and cluster-level artifact audits.

The qualitative error taxonomy in Table 36 states which failure cases the framework is designed to expose. The taxonomy is diagnostic rather than a measured error distribution unless a released artifact contains per-category annotations and adjudication logs. This distinction matters because some categories, such as version boundaries, require human judgment beyond metadata-derived silver evidence.

Table 36: Error taxonomy for identity-agenda confusion. The table identifies the expected failure mode, the IAD-Risk control, and the audit evidence needed for stronger claims.

Error category	Typical source of confusion	IAD-Risk control	Stronger audit evidence
Same task, different contribution	Papers share a benchmark, model family, or problem statement.	Agenda-non-identity head and HNFMR reporting.	Human-reviewed hard-negative slice with pair notes.
Citation-neighborhood neighbor	Papers share references or citation community but are distinct works.	Reference-overlap features contribute to agenda evidence, not direct merge evidence.	Citation-derived pair provenance and checksum-fixed predictions.
Version or extension boundary	Preprint, short paper, conference, and journal records are similar but may not be mergeable.	Conservative defer/block behavior when identity evidence is incomplete.	Manual adjudication with version-policy labels.
Identifier conflict	DOI, arXiv, or OpenAlex identifiers disagree despite text similarity.	Different-identifier conflicts enter the ANI risk head and cannot-link logic.	Identifier-resolution logs and conflict audit table.
Sparse metadata	Abstract, venue, author, or identifier fields are missing or noisy.	Merge requires positive identity evidence rather than semantic similarity alone.	Missing-field stratified metrics and per-source diagnostics.

The error analysis emphasizes claim discipline. The current evidence supports identity-agenda risk modeling, stratified benchmark construction, and targeted false-merge suppression. It does not support claims that OpenAlex labels are human gold, that the system has broad domain superiority, that component-level ablations have already closed every mechanism question, or that manual annotation has already been completed.

8. Limitations

This study has five limitations. First, OpenAlex/OpenCitations hard negatives are silver labels, not human gold. They are useful for stress testing but must be reported separately from DeepMatcher gold labels. Second, SciRepEval/SciDocs-style proximity is a proxy for agenda relatedness, not proof of identity or non-identity. Third, source-heldout generalization remains mixed across model variants and should be expanded before making broad claims across scholarly domains. Fourth, pair-level metrics do not by themselves establish cluster-level deployment quality; cluster assignments, cannot-link coverage, and cluster contamination rate are required before making cluster-level quality claims. Fifth, an independently reviewed 500–1,000 pair validation slice would provide stronger evidence about label precision and difficult boundary cases; the supplementary material specifies the sampling, blinding, adjudication, and agreement-reporting protocol required before making stronger human-validation claims.

9. Threats to Validity

Construct validity depends on whether the label strata measure the intended relations. DeepMatcher pairs provide same-work identity evidence, but OpenAlex/OpenCitations hard negatives are silver stress-test pairs rather than manual non-identity gold. Internal validity depends on threshold selection and split separation; therefore threshold-sensitive systems must select thresholds on validation data and report held-out metrics. External validity is limited by the current source mix: the Open-v2 evidence supports the identity-agenda mechanism, while broader claims require more source-heldout and domain-diverse evaluation. Reproducibility depends on public data availability and stable processing scripts because the repository deliberately avoids redistributing large third-party data files.

Table 37 summarizes the main validity risks and the safeguards used in the manuscript. The table is intentionally conservative: a mitigation reduces a specific validity risk, but it does not turn proxy or silver evidence into human-adjudicated truth.

Table 37: Threats to validity and claim boundaries. Each risk is paired with the current mitigation and the remaining statement that should not be over-claimed.

Validity dimension	Main risk	Current mitigation	Remaining boundary
Construct validity	Silver hard negatives may contain metadata noise.	Label strength is preserved; silver rows remain stress-test evidence.	Not a replacement for manual non-identity gold.
Internal validity	Threshold choice or split leakage can inflate metrics.	Operating points, split fields, and pair-leakage audits are required.	Numerical claims need released predictions and logs.
External validity	Open-v2 may not cover every source, topic, or version policy.	Claims are tied to Open-v2 and source-heldout evidence boundaries.	Broad deployment requires domain-diverse validation.
Conclusion validity	HNFMER evidence may be mistaken for full causality.	Ablation requirements precede component-level causal claims.	Complete causality requires no-gate, no-ANI, and single-space variants.
Reproducibility	Raw data and large predictions are not stored in Git.	Public commands, fixtures, manifests, checksums, and artifact rules are provided.	Full numeric audit requires L2/L3 artifacts.
Operational validity	Pair errors can propagate through transitive merges.	Risk gating and cannot-link-aware clustering are described.	Cluster guarantees need complete cannot-link coverage.

10. Claim Interpretation Boundary

Table 38 summarizes the claim boundary that should guide interpretation of the manuscript. The boundary table makes the main evidence location, supported wording, and boundary before stronger wording explicit for readers. It also separates manuscript-supported conclusions from claims that require artifact-backed ablations, a manual-validation slice, or broader source-heldout validation.

Table 38: Claim interpretation boundary. Each row maps a common review dimension to the present evidence and the wording boundary that should not be crossed without additional artifacts.

Review dimension	Main evidence location	Supported wording	Boundary before stronger wording
Contribution clarity	Introduction, problem formulation, and IAD-Bench contract.	The manuscript identifies identity-agenda confusion and HNFMR as a false-merge safety problem.	Do not frame the work as a universal replacement for entity resolution or scientific representation learning.
Writing reproducibility	Method, implementation details, data/code availability, and supplementary protocol.	The repository supports code-level checks, fixture rebuilds, schema validation, and artifact-release preparation.	Full numerical audit requires external artifacts with predictions, logs, manifests, and checksums.
Experimental strength	Open-v2 evidence snapshot, operating-point disclosure, and result audit trail.	Current evidence supports targeted false-merge suppression under the stated Open-v2 scope.	Stronger ranking claims require same-scope predictions, threshold logs, bootstrap intervals, and released metric tables.
Evaluation completeness	Claim-evidence boundary, manual-validation boundary, and uncertainty requirements.	The manuscript separates gold, proxy, silver, and future manual-review evidence layers.	Component causality or human-gold claims require artifact-backed ablations, threshold grids, and a manual-validation slice.
Method design soundness	Risk score rationale, merge-decision rule, and validity threats.	IAD-Risk exposes identity, agenda, ANI risk, thresholds, and cannot-link behavior.	Broad deployment claims require source-heldout validation, topic-heldout checks, and failure-case analysis.

11. Conclusion

IAD-Risk addresses a specific failure mode in scholarly entity matching: the tendency to confuse agenda similarity with work identity. The core idea is to separate identity and agenda evidence before a pair is allowed to become an automatic merge. By adding agenda-non-identity supervision and a false-merge risk gate, the framework turns scholarly deduplication from an unconditional similarity decision into a selective decision process that can block or defer risky pairs.

The Open-v2 evidence supports a conservative conclusion: identity-agenda disentanglement can provide targeted false-merge suppression under the reported pair scope and operating points. IAD-Bench contributes a reproducible benchmark contract by keeping gold identity labels, proxy agenda evidence, and silver hard negatives separate throughout training, evaluation, and claim interpretation. This separation is the main practical lesson of the study: relatedness evidence is useful for retrieval and candidate generation, but it must not be treated as sufficient evidence of work identity.

The manuscript does not claim cluster-level deployment quality without cluster artifacts, nor does it claim broad method ranking across scholarly domains. Additional validation should expand independent manual review, source-heldout evaluation, same-scope baseline prediction releases, and cluster-level artifact audits. These extensions would make it possible to test whether the present pair-level safety mechanism transfers to broader source mixtures and production-style deduplication workflows.

Data and Code Availability

The source code, benchmark construction scripts, public-release checks, and manuscript build scripts are organized in the project repository. After installation, the reproducibility entry points include CLI commands for DeepMatcher conversion, proximity conversion in the SciRepEval style, weak-label construction from OpenAlex and OpenCitations, IAD-Bench assembly, stratification checks, source-bias diagnostics, baseline execution, IAD-Risk training, bootstrap analysis, and manuscript PDF generation.

Raw third-party data are not redistributed in the repository because the original sources have their own access conditions and licenses. The repository instead provides data-processing commands, schema contracts, fixture tests, and artifact-release guidance so that independent readers can reproduce the

benchmark construction from public sources and verify generated artifacts through manifests, checksums, command logs, and fixed commit identifiers.

Table 39: Data and code availability boundary. The table separates what is version-controlled in Git from what must be obtained from public sources or a released artifact package.

Availability class	Included location	Evaluation use
Source code and CLI entry points	Project repository.	Supports installation, schema checks, fixture rebuilds, baseline execution, model training entry points, and manuscript builds.
Small public fixtures and schema contracts	Project repository.	Supports L0/L1 code-level reproduction without downloading full raw datasets.
Raw third-party source files	Excluded from Git; obtained from original providers.	Raw third-party data remain governed by original provider licenses and access conditions.
Full prediction files and model checkpoints	Excluded from Git unless provided through an external artifact release.	Prevents large local outputs from being treated as repository source files.
Derived evaluation artifacts	External artifact release.	Supports L2/L3 result-level audit through prediction files, threshold logs, manifests, checksums, and commit identifiers.

This boundary distinguishes L0/L1 code-level reproduction from L2/L3 result-level audit. Full numerical reproduction requires public-source rebuilds or released artifacts; the Git repository alone verifies the executable contracts, fixture behavior, and manuscript build rather than redistributing raw data or full experiment outputs.

Ethics Statement

This study uses public scholarly metadata and benchmark-style record linkage data. It does not involve human participants, clinical records, private user behavior, or animal experiments. Any future manual validation should be limited to scholarly record relation labels and should avoid collecting sensitive personal information.

Competing Interests

The authors declare no competing interests.

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